

ZS-F28

Seam-Constrained Finite Fourier Analysis II: The Symmetrized Lam–Leung Theorem, Its Equivariant Extension, and the Purity Frontier

Kenny Kang Z-Spin Collaboration · March 2026 Theme: Math Spine / Seam-Constrained Programme · Paper code: ZS-F28 · v1.3 (companion to ZS-F27 v1.3, §3.5–§3.7; supersedes ZS-F28 v1.0, v1.1, v1.2)

Verification: 45/45 PASS | Zero Free Parameters | NO CLAIM on unrestricted Fuglede | Purity Conjecture now PROVEN for the entire $\min(a,b)=1$ family (Theorem B); OPEN only for two-square groups $\min(a,b)\geq 2$, where the naive T1-formulation is shown FALSE and the spectral formulation is the correct one

External-submission title (Z-Spin-free version; see §8.4): *An Equivariant Lam–Leung Theorem for Two-Prime Cyclic Groups, with Applications to Fuglede's Conjecture for Symmetric Sets.*

§0. Abstract

This release advances the seam-restricted Fuglede programme from a partially-verified conjecture to a proven theorem on an infinite family, and in doing so discovers the precise frontier where the naive formulation fails. All results are stated and proved without Z-Spin vocabulary (§3–§7; the corpus interface is confined to §11 and deleted in the external-submission version, §8.4). **The headline: a purity characterization (Theorem A, PROVEN).** For $d = p^a q$ with one prime squarefree, a vanishing set $A \subseteq \mathbb{Z}_d$ admits an all- p -cycle (pure) decomposition **if and only if** $\Phi_{\{p^a\}} \mid A(x)$. The proof is a one-line consequence of the product structure of q -cycles ($A_q = U_q \times \mathbb{Z}_q$), and it reduces single-direction purity — the engine of the descent — to a single cyclotomic divisibility. Verified across 8 two-prime groups in both prime orderings (480 000 vanishing sets, zero equivalence violations). This supersedes the v1.2 size argument (which reached only $a = 2$, $p < q$, proper subsets) and removes every restriction for the squarefree-exponent family. **The consequence (Theorem B, PROVEN): the Purity Conjecture holds for every two-prime group with $\min(a,b) = 1$.** Combining Theorem A on A with the same theorem on the tiling complement, no genuinely mixed seam-invariant set is a tile, hence (by Malikiosis–Kolountzakis [3]) none is spectral; equivalently every symmetric spectral set in $\mathbb{Z}_{\{p^a q\}}$ or $\mathbb{Z}_{\{p q^b\}}$ admits a pure decomposition. For these groups the two formulations of the conjecture — "genuinely mixed $\Rightarrow$ fails Coven–Meyerowitz T1" and "symmetric spectral $\Rightarrow$ admits a pure decomposition" — coincide and are both proven. Exhaustive at $d = 45, 63, 75$ (0 genuinely-mixed tiles out of 96, 336, 6000). **The discovery (the $p^2 q^2$ divergence).** At the minimal two-square group $d = 225 = 3^2 \cdot 5^2$ the two formulations **separate**: the T1-formulation is **false** — we exhibit explicit seam-invariant genuinely-mixed vanishing sets satisfying T1 (an explicit witness of size 75 with cyclotomic divisor set $\{\Phi_5, \Phi_9, \Phi_{25}\}$, T1-admissible cardinality $3 \cdot 5 \cdot 5 = 75$), all of which fail Coven–Meyerowitz T2 (the witness lacks Φ_{45}) and are therefore non-spectral. Across roughly $6 \cdot 10^5$ genuinely-mixed sets tested at $d = 225$, **148 satisfy T1 yet not one is spectral.** The single clean structural obstruction that survives is at

cardinality 25, where Φ_5 and Φ_{25} are provably incompatible (Φ_{25} forces the five p-cycles onto one order-5 coset, after which Φ_5 would require 12 q-cycles where only 2 are available; exhaustively confirmed over all 1215 Φ_{25} -structured configurations). **Upshot for the programme.** The Purity Conjecture is now correctly stated in its spectral (T2-using) form, PROVEN for the infinite family $\min(a,b) = 1$, and OPEN only for two-square groups $\min(a,b) \geq 2$, where the naive T1-form is refuted. This is exactly why the spectral $\Rightarrow$ T1 question (does spectrality force T1 without passing through T2?) is the real branch point O-F28.5, sharpened here from a programmatic note to a theorem-level dichotomy. The earlier reductions are preserved: the pure-layer descent (Proposition 6.1), the honest re-scoping of the one-level descent corollary as a consistency demonstration inside the region [3] already settles (Corollary 6.5), and the cheap T2-falsification showing seam symmetry alone does not force T2 (Proposition 7.1). Closed forms $a(pq) = 2^{\{(p+1)/2\}} + 2^{\{(q+1)/2\}} - 3$ and $a(p^a) = 2^{\{(p^{a-1}+1)/2\}} - 1$ (PROVEN; OEIS-prepared, Appendix D). Verification: 45/45 checks (zs_f28_verify_v1_3.py, exact integer cyclotomic arithmetic; exact maximum-clique spectral testing; seeds 20260612, 20260613), including Theorem A over 8 groups (Y1), the size-25 no-go (Y2), and the explicit p^2q^2 divergence witness (Y3).

§0.1 Epistemic Status Legend

TAG	MEANING
PROVEN	Rigorous proof under declared definitions; verified by direct computation where finite; no open step.
DERIVED	Valid deduction from PROVEN results and imports; zero new free parameters.
DERIVED-CONDITIONAL	DERIVED conditional on an explicitly listed open statement.
IMPORTED-PROVEN	External peer-reviewed or community-verified result used without re-proof; full citation.
COMPUTED	Exact finite computation (exhaustive enumeration or pre-registered sampling); evidence, and where exhaustive, proof of the finite statement.
HYPOTHESIS-strong	Motivated conjecture with explicit closure path and structural anchors; not derived.
OPEN	Registered open problem.
NON-CLAIM	Explicitly not asserted.
OBSERVATION	Structural coincidence registered without mechanism; no evidentiary weight.

§1. Introduction

ZS-F28 v1.0 closed Step 3 of the seam-restricted Fuglede programme registered in ZS-F27 §3.7: it retired the conjectured single-direction collapse by exhaustive falsification at $d = 45$ and replaced it with the Symmetrized Lam–Leung Theorem (Theorem 4.3), proved twice — by reflection across seam-orbit representatives of the class lattice, and by a gauge argument whose positivity reduces to two interval identities. Two independent audit reports on v1.0 then converged on the same diagnosis from different directions: the theorem is correct and useful but its external legibility and its forward force were both under-developed. The first report prescribed an equivariant generalization of the theorem, a one-level descent target, exhaustive computation beyond $d = 45$, a set-level (0/1) strengthening, an explicit novelty-scope sentence, and an external title. The second report identified a missing elementary fact with disproportionate consequences — that on odd d the seam is the translation conjugate of negation, so the entire programme has an external name (*Fuglede for symmetric, i.e. real-mask, sets*) and a correct keyword set for literature audit — and prescribed a cheap computational falsification of the implicit hope that symmetry forces the Coven–Meyerowitz conditions, plus the registration of the enumeration data itself as a citable artifact.

Version 1.1 executes both prescriptions and reports what the executions returned. The repositioning is now Proposition 4.7 / Corollary 4.8, and it came with a precise boundary: the conjugacy holds exactly when d is odd, so the seam is a genuinely separate symmetry class exactly when $p = 2$ is admitted — the equivariant Theorem 4.9 covers both classes, exhaustively machine-verified on even d . The cheap falsification was run and returned a clean negative (Proposition 7.1): symmetry alone does not force T2 — killing one route cheaply, as designed. The descent target was reached in the strongest form the evidence supports: an unconditional descent mechanism (Proposition 6.1), a proven purity obstruction at $\mathbb{Z}_{p^2q}$ (the v1.1 size argument, since superseded by Theorem A, §6.2), a one-level descent in the known region (Corollary 6.5), and — from two new exhaustive enumerations at $d = 63$ and $d = 75$, the latter chosen precisely because the size obstruction is silent there — the Purity Conjecture (Conjecture 6.4), which, if proven, deletes the mixed obstruction from the main target's induction wherever spectral $\Rightarrow$ T1 is available. The set-level strengthening requested by the first report is absorbed into this purity architecture, which is its sharper form: the question "what does symmetric decomposition force on 0/1 supports?" becomes "do mixed supports ever satisfy T1?", and the answer so far, over 6432 exhaustively enumerated mixed sets across three group sizes, is never.

The paper remains organized external-first: §3–§7 are plain finite Fourier analysis on cyclic groups, and a reader with no knowledge of the Z-Spin corpus loses nothing before §11. On novelty, the scope is fixed here as the first report prescribed: we do not claim priority over all equivariant refinements of Lam–Leung-type decompositions; the novelty claim is restricted to the explicit reflection family $\tau_r(x) = r - x$ on two-prime cyclic groups, the constructive symmetric cycle decomposition, its translation normal form and even- d boundary, and its use as the engine of the symmetry-restricted Fuglede programme (NC-F28.3, NC-F28.5).

v1.3 advances. Where v1.1–v1.2 verified the Purity Conjecture and reduced the programme to it, v1.3 *proves* it for an infinite family and locates exactly where it breaks. Theorem A (§6.2) characterizes

single-direction purity by one cyclotomic divisibility, $\Phi_{\{p^a\}} \mid A$, for every group with a squarefree exponent; Theorem B (§6.3) parlays this, through the tiling complement, into a full proof of the Purity Conjecture for the entire $\min(a,b) = 1$ family — the first unconditional closure of the programme's forward step on infinitely many groups. The price of honesty is the §6.4 discovery: at the minimal two-square group $d = 225$ the conjecture's two equivalent-looking formulations diverge, and the combinatorial "fails T1" form is false (explicit witness), so the live statement is the spectral form and the live obstruction is T2 — which is precisely the O-F28.5 branch point, now upgraded from a programmatic note to a proven dichotomy. None of this touches the unrestricted Fuglede conjecture (NC-F28.1).

v1.2 corrections (retained). A third audit, on v1.1, found two local defects in §6, both repaired in v1.2 and carried forward here. (1) The v1.1 Proposition 6.2 asserted that a T1-admissible-cardinality vanishing set has *only pure* decompositions; this is false — $\mathbb{Z}_{\{p^2q\}}$ itself has T1-admissible cardinality p^2q yet admits a mixed decomposition (explicit witness, §6.2, check X1). The corrected proposition restricts to proper subsets and concludes "not genuinely mixed" (existence of a pure decomposition), which is all the descent uses; the general $p < q$ proof, previously delegated to three exhaustive instances, is now an explicit divisor argument written into the proof body. (2) The v1.1 Corollary 6.5 claimed to recover the $\mathbb{Z}_{\{p^2q\}}$ case "by the purity route rather than by importing"; since $\text{spectral} \Rightarrow \text{T1}$ on $\mathbb{Z}_{\{p^nq\}}$ is itself the content of the imported proof [3], this over-claimed, and the corollary is re-labelled a one-level consistency demonstration of the descent template. The repair is productive: it isolates the programme's true bottleneck as the question whether $\text{spectral} \Rightarrow \text{T1}$ holds unconditionally on general $\mathbb{Z}_{\{p^mq^n\}}$, now registered as O-F28.5 and made the center of the sharpened expert query (§8.4). The audit also asked for the minimal two-square group $d = 225$ to be pushed; §6.4 reports a structured slice there (3712 mixed sets, none spectral or T1).

§2. Locked Inputs and Imports

Table 2.1. Inputs and imports.

Object	Statement	Status
Seam involution	$\tau(x) = -1-x = N-1-x$ on $\mathbb{Z}_N$; equals $J_N = F_{N^2}S$ (ZS-F27 Theorem 3.3/3.4).	LOCKED (corpus) / canonical
Half-angle pinning	Seam-invariant masks have real centered Fourier values (ZS-F27 Lemma 3.11).	IMPORTED-PROVEN (internal)
Quotient stability	Seam-invariance is preserved under every quotient $\mathbb{Z}_N \rightarrow \mathbb{Z}_M$ (ZS-F27 Lemma 3.13).	IMPORTED-PROVEN (internal)
Pilot closure	Seam-invariant Fuglede equivalence on $\mathbb{Z}_{\{p^nq\}}$, by restriction of [3] (ZS-F27 Corollary 3.12).	IMPORTED (internal, DERIVED)

Object	Statement	Status
Two-prime vanishing sums	Every vanishing $\mathbb{N}$ -sum of d -th roots, $d = p^a q^b$, is an $\mathbb{N}$ -combination of rotated p -cycles and q -cycles.	IMPORTED-PROVEN [1, 2]
Fuglede on $\mathbb{Z}_{\{p^n q\}}$	Spectral $\Leftrightarrow$ tile [3].	IMPORTED-PROVEN
Fuglede on $\mathbb{Z}_{\{p^n q^2\}}$	Spectral $\Leftrightarrow$ tile, via the Fuglede–Pompeiu connection [4].	IMPORTED-PROVEN
Conditional $\mathbb{Z}_{\{p^m q^n\}}$	Spectral $\Rightarrow$ tile when an exponent is ≤ 6 or $p^{m-2} < q^4$ [5].	IMPORTED-PROVEN
Tile $\Rightarrow$ Spectral on $\mathbb{Z}_{\{p^m q^n\}}$	Holds for all m, n [6]; with [7] ($T1 \wedge T2 \Rightarrow$ spectral).	IMPORTED-PROVEN
Coven–Meyerowitz conditions	Cyclotomic tiling conditions $T1, T2$; $T1$ necessary for all tiles; $T1 \wedge T2$ sufficient; both necessary when $ A $ has ≤ 2 prime factors [8].	IMPORTED-PROVEN
Symmetry precedent	Spectral convex bodies in $\mathbb{R}^d$ are centrally symmetric [15]: symmetry restrictions are an established structure class in the Fuglede problem family.	IMPORTED-PROVEN (context)
Three-prime context	$\mathbb{Z}_{\{pqr\}}$ [9], $\mathbb{Z}_{\{p^2 qr\}}$ [10], $\mathbb{Z}_{\{p^n qr\}}$ [11], $\mathbb{Z}_{\{pqrs\}}$ [12].	IMPORTED (context)

§3. Setup

Fix distinct primes p, q and $d = p^a q^b$ ($a, b \geq 1$). By the Chinese remainder theorem identify $\mathbb{Z}_d \cong \mathbb{Z}_{\{p^a\}} \times \mathbb{Z}_{\{q^b\}}$, writing elements (u, v) . A **p -cycle** is a coset of the unique order- p subgroup, $\{(u_0 + k p^{a-1}, v_0) : 0 \leq k < p\}$; its index is the pair $(u_0 \bmod p^{a-1}, v_0)$. A **q -cycle** is defined symmetrically. A **cycle decomposition** of an $\mathbb{N}$ -multiset $m: \mathbb{Z}_d \rightarrow \mathbb{N}$ is an $\mathbb{N}$ -weighting of cycles summing to m ; in coordinates this is the separable form

$$m(u, v) = w(u \bmod p^{a-1}, v) + y(u, v \bmod q^{b-1}), \quad w, y \geq 0.$$

The **class lattice** is $C = \mathbb{Z}_{\{p^{a-1}\}} \times \mathbb{Z}_{\{q^{b-1}\}}$. New in v1.1: the **reflection family** is $\tau_r(x) = r - x$ for $r \in \mathbb{Z}_d$; the seam is $\tau = \tau_{-1}$. Every τ_r reduces componentwise under CRT and under every quotient (r reduces to r), maps p -cycles to p -cycles and q -cycles to q -cycles through the index maps, and induces the involution $\sigma_r(\gamma) = r - \gamma$ componentwise on C ; τ_r -invariance of a multiset is therefore well-posed at every scale, exactly as for the seam (ZS-F27 Lemma 3.13 with -1 replaced by r). By the imported structure theorem [1, 2], an $\mathbb{N}$ -multiset m satisfies $\sum m(x) \zeta_d^x = 0$ if and only if it admits a cycle decomposition; for a 0/1 multiset (a set), positivity forces the chosen cycles to be disjoint, so a

vanishing **set** is exactly a disjoint union of p-cycles and q-cycles. The **gauge freedom** of a decomposition is $w \mapsto w - g, y \mapsto y + g$ for integer class functions $g: C \rightarrow \mathbb{Z}$.

§4. The Symmetrized Lam–Leung Theorem and Its Equivariant Extension

§4.1–§4.6 reproduce v1.0 (restated for self-containment); §4.7–§4.10 are new in v1.1.

Lemma 4.1 (defect identities). [PROVEN] Let m be seam-invariant and (w, y) any integer decomposition. Define $\delta_w(c, v) = w(c, v) - w(-1-c, -1-v)$ and $\delta_y(u, v') = y(-1-u, -1-v') - y(u, v')$. Then: (i) both defects descend to one and the same class function $\delta: C \rightarrow \mathbb{Z}$; (ii) $\delta \circ \sigma = -\delta$; (iii) δ vanishes on every σ -fixed class. *Proof.* Expanding $0 = m(u, v) - m(\tau(u, v))$ in the separable form yields $\delta_w(c, v) = \delta_y(u, v')$ for every compatible pair; the left side is independent of u and the right of v , giving (i); (ii) by evaluating at reflected arguments; (iii) since $\delta(\gamma) = -\delta(\gamma)$ over $\mathbb{Z}$. $\square$ (Verified: check F1, all 379 sets at $d = 45$.)

Lemma 4.2 (class-interval characterization). [PROVEN] With $L(\gamma) = -\min \{ y(u, v') : \text{over } \gamma \}$ and $U(\gamma) = \min \{ w(c, v) : \text{over } \gamma \}$, m admits an $\mathbb{N}$ -decomposition iff $L \leq U$ on every class, and the $\mathbb{N}$ -decompositions are exactly the gauges $L \leq g \leq U$. (Verified: within check F2.)

Theorem 4.3 (Symmetrized Lam–Leung Theorem). [PROVEN] Let $d = p^a q^b$ and let $m: \mathbb{Z}_d \rightarrow \mathbb{N}$ be seam-invariant. If m admits a cycle decomposition, it admits a seam-symmetric cycle decomposition: $w \circ \sigma = w$ and $y \circ \sigma = y$. *Proof (reflection construction).* Take any $\mathbb{N}$ -decomposition (w, y) . Choose a set R of representatives of the σ -orbits on C ; leave the values over classes in R unchanged and overwrite the values over every non-representative class by the seam-reflected values. Gluing over the overwritten classes is precisely the seam-invariance of m ; symmetry over fixed classes is the vanishing of the defect, Lemma 4.1(iii); every value of the new pair is a value of the old, so $\mathbb{N}$ -positivity is inherited. $\square$ *Alternative proof (gauge).* Solve $g - g \circ \sigma = \delta$ (consistent by Lemma 4.1(ii)–(iii)); the symmetric solution is feasible in $\mathbb{N}$ because $U(\sigma\gamma) + \delta(\gamma) = U(\gamma)$ and $L(\sigma\gamma) + \delta(\gamma) = L(\gamma)$, so the constrained interval collapses to $[L(\gamma), U(\gamma)]$, nonempty whenever any $\mathbb{N}$ -decomposition exists. $\square$ (Verified: check E1 — reflection construction executed on all 379 exhaustive sets at $d = 45$; E1b — symmetric exact covers on all 379; E2–E3 — 500 + 1300 sampled multisets at $d = 45$ and $d = 225$, seeds 20260612/20260613; F2 — interval identities on all exhaustive instances.)

Corollary 4.4 (squarefree degeneracy). [PROVEN] If $a = b = 1$ the class lattice is a single σ -fixed point, so $\delta \equiv 0$: every $\mathbb{N}$ -decomposition of a seam-invariant multiset of $\mathbb{Z}_{pq}$ is automatically seam-symmetric. (Verified: check C2.)

Corollary 4.5. [DERIVED from IMPORTED-PROVEN] Every seam-invariant $\mathbb{N}$ -multiset of $\mathbb{Z}_{p^a q^b}$ with $\sum m(x) \zeta_d^x = 0$ admits a seam-symmetric decomposition into p-cycles and q-cycles. (Theorem 4.3 $\circ [1, 2]$.)

Corollary 4.6 (sets at the squarefree top level). [PROVEN] A seam-invariant subset $A \subseteq \mathbb{Z}_{pq}$ with $A(\zeta_{pq}) = 0$ is a disjoint union of fibers in a single direction, and its fiber index set is seam-invariant in

the quotient. (Verified: checks B1–B2; squarefree mixed counts 0 at $d = 21, 33, 35$ — check P2.) This is the honest content behind the original "single-direction collapse" intuition — a squarefree-level phenomenon, and §5 shows it stops there.

§4.7 Translation normal form of the seam (new).

The second audit report observed that the seam is not an exotic reflection but the translation conjugate of negation on odd groups. We record this precisely, with the even- d boundary made explicit, because it fixes the programme's external name and keyword set.

Proposition 4.7 (reflection normal form). [PROVEN] On $\mathbb{Z}_d$ the affine involutions are exactly the reflections $\tau_r(x) = r - x$ ($r \in \mathbb{Z}_d$), together with the identity. Two reflections τ_r, τ_s are translation-conjugate ($\tau_s = T_{\{-t\}} \circ \tau_r \circ T_t$ with $T_t(x) = x + t$) iff $s \equiv r + 2t \pmod{d}$. Hence: (i) if d is **odd**, 2 is invertible mod d and all reflections form a **single** conjugacy class; in particular the seam $\tau_{\{-1\}}$ is conjugate to negation $\tau_0 = (x \mapsto -x)$ by T_t with $2t \equiv -1$, i.e. $t = (d-1)/2 \equiv -(d+1)/2$; (ii) if d is **even**, $r \mapsto r + 2t$ preserves the parity of r , so there are **exactly two** classes — the "even reflections" (r even, two fixed points, conjugate to negation τ_0) and the "odd reflections" (r odd, no fixed point, conjugate to the seam $\tau_{\{-1\}}$ since -1 is odd). *Proof.* $\tau_s = T_{\{-t\}} \tau_r T_t$ sends $x \mapsto r - (x + t) - (-t) \dots$ computing, $T_{\{-t\}} \tau_r T_t(x) = (r - (x+t)) - t = (r - 2t) - x = \tau_{\{r-2t\}}(x)$; so conjugation shifts the reflection parameter by $-2t$, and the orbit of r under $r \mapsto r - 2t$ is $r + 2\mathbb{Z}_d$. For odd d this is all of $\mathbb{Z}_d$ (one class); for even d it is the parity coset of r (two classes). The fixed points of τ_r solve $2x \equiv r$: two solutions when r is even, none when r is odd (d even); one solution when d is odd. $\square$ (Verified: check N1 — explicit conjugacy seam $\leftrightarrow$ negation at $d = 15, 45, 105$; check N2 — reflection-class transport of the vanishing count, all r giving 379 at $d = 45$.)

Corollary 4.8 (real-mask description of seam-invariant sets). [PROVEN] Let d be odd and $A \subseteq \mathbb{Z}_d$. Then A is seam-invariant iff its recentering $A' = A + (d+1)/2$ is negation-invariant ($A' = -A'$), iff the mask $A'(x)$ is **self-reciprocal**, $A'(x) \equiv x^{\wedge\{d\}} A'(x^{-1}) \equiv A'(x^{-1})$ on the d -th roots, iff $A'(\zeta) \in \mathbb{R}$ for every d -th root ζ . Thus, on odd groups, the seam-restricted Fuglede programme is exactly **Fuglede's conjecture for sets with real mask polynomial** (equivalently, negation-symmetric / centrally symmetric sets), a translation-invariant subclass. *Proof.* Translation by $t = (d+1)/2$ conjugates $\tau_{\{-1\}}$ to τ_0 (Proposition 4.7(i)); negation-invariance of A' is the reality of $A'(\zeta) = \sum_{a \in A'} \zeta^{\wedge a} = \sum \cos$ for the cosine pairing $a \leftrightarrow -a$; self-reciprocity is the polynomial form of the same statement. This is the recentered form of the half-angle pinning ZS-F27 Lemma 3.11. $\square$ (Verified: check N3 — A seam-invariant $\Rightarrow A + (d+1)/2$ negation-invariant, all 379 sets at $d = 45$.)

The consequence the audit emphasized: the correct external keywords are "symmetric sets", "real vanishing sums of roots of unity", "self-reciprocal mask polynomials", and "Fuglede for centrally symmetric sets" — not "seam". Under these keywords the literature precedent is real and supportive: spectral convex bodies must be centrally symmetric [15], so symmetry is an established and load-bearing structure class in the Fuglede family, not an idiosyncratic constraint. The boundary in Proposition 4.7(ii) is equally important: on even d the seam (odd reflection, fixed-point-free) is **not** translation-equivalent to

negation, so the seam genuinely names a distinct symmetry only when the prime 2 participates — which is exactly the regime the equivariant theorem below is built to cover.

§4.8 The equivariant theorem (new).

Theorem 4.9 (Equivariant Lam–Leung Symmetrization). [PROVEN] Let $d = p^a q^b$ (p, q distinct primes, **2 admitted**), let $r \in \mathbb{Z}_d$, and let $\tau_r(x) = r - x$ be the corresponding affine reflection. Every τ_r -invariant $\mathbb{N}$ -multiset $m: \mathbb{Z}_d \rightarrow \mathbb{N}$ admitting a cycle decomposition admits a τ_r -**symmetric** cycle decomposition ($w \circ \sigma_r = w, y \circ \sigma_r = y, \sigma_r(\gamma) = r - \gamma$ on the class lattice). *Proof.* The reflection and gauge proofs of Theorem 4.3 use only three facts about the involution, each of which holds verbatim for every τ_r and every d : (1) τ_r maps p -cycles to p -cycles and q -cycles to q -cycles through the index maps (because translation and negation each preserve the order- p and order- q subgroups, and r reduces componentwise under CRT); (2) the defect of any decomposition descends to a class function on C , antisymmetric under σ_r and zero on σ_r -fixed classes (the master equation $\delta_w = \delta_y$ of Lemma 4.1, with $\tau_{\{-1\}}$ replaced by τ_r , is unchanged); (3) the interval identities $U \circ \sigma_r + \delta = U, L \circ \sigma_r + \delta = L$ hold by the same one-line computation. The reflection construction overwrites non-representative σ_r -orbits by τ_r -reflected values; gluing is τ_r -invariance of m ; positivity is value inheritance. Nothing used the oddness of d or the value -1 of r . $\square$ (Verified, exhaustive, both reflection classes: check N4 — vanishing-set counts at $d = 12, 20, 24, 36$ for $r = 0$ (negation/even class) and $r = -1$ (seam/odd class): $\{23, 9\}, \{79, 33\}, \{239, 99\}, \{2399, 999\}$ respectively; check N5 — symmetric exact covers and the reflection construction valid on every one of those instances; check N6 — exhaustive multiset verification at $d = 12$ cap 2, 170/170 for $r = 0$ and 44/44 for $r = -1$; check E4 — 500 sampled multisets at $d = 72 = 2^3 \cdot 3^2$.)

Theorem 4.9 is the form to submit externally: it makes no reference to a distinguished involution, covers $p = 2$, and reduces to Theorem 4.3 at (d odd, $r = -1$). It answers the first audit's "equivariant Lam–Leung theorem" request and the second audit's " $p = 2$ boundary" request in one statement.

§4.9 Squarefree and prime-power enumeration closed forms (new).

The exhaustive enumerations produce two clean counting laws, recorded both as structural facts and as the seed rows of the OEIS artifact (Appendix D).

Proposition 4.10 (enumeration closed forms). [PROVEN] Let $a(d)$ be the number of nonempty seam-invariant subsets $A \subseteq \mathbb{Z}_d$ with $A(\zeta_d) = 0$ (equivalently $\Phi_d \mid A(x)$). (i) **Squarefree:** $a(pq) = 2^{\{(p+1)/2\}} + 2^{\{(q+1)/2\}} - 3$ for distinct odd primes p, q . (ii) **Prime power:** $a(p^a) = 2^{\{(p^a-1)/2\}} - 1$. *Proof sketch.* By Corollary 4.6 every vanishing seam-invariant set of $\mathbb{Z}_{\{pq\}}$ is a single-direction fiber union with seam-invariant index set in the quotient; the p -direction unions are indexed by nonempty negation-symmetric subsets of $\mathbb{Z}_q$ (count $2^{\{(q+1)/2\}} - 1$, since the $(q-1)/2$ reflection pairs plus the center give a free binary choice on $(q+1)/2$ orbits, minus the empty set) and symmetrically for q ; the two families meet only in the empty union, giving inclusion–exclusion $2^{\{(p+1)/2\}} + 2^{\{(q+1)/2\}} - 1 - 2$ (the full-group set is counted once in each and subtracted). For prime powers, a vanishing seam-invariant set is a union of order- p fibers whose index set in $\mathbb{Z}_{\{p^a-1\}}$ is seam-invariant; the count of nonempty

such index sets is $2^{\{(p^{a-1}+1)/2\}} - 1$. $\square$ (Verified: check P1 at $d = 15, 21, 33, 35 \rightarrow 9, 17, 65, 21$; check P3 at $d = 9, 25, 27, 49 \rightarrow 3, 7, 31, 15$.)

These match the v1.0 hand count $a(15) = 9$ and supply the previously unproven $a(45)$ context: 379 is *not* of this closed form because $45 = p^2q$ is the first genuinely mixed group, which is the whole point of §5–§6.

§5. Negative Result: the Single-Direction Form is False (unchanged)

Proposition 5.1. [PROVEN by exhaustive computation] At $d = p^2q = 45$ the single-direction collapse fails: of the 379 seam-invariant subsets of $\mathbb{Z}_{45}$ with $\Phi_{45} \mid A(x)$ (the complete list, by exact integer cyclotomic arithmetic with a meet-in-the-middle search over the 23 seam orbits), exactly 96 are **genuinely mixed** — no $\mathbb{N}$ -decomposition using only p -cycles, and none using only q -cycles. Smallest explicit example (size 11): $A = \{0, 4, 13, 14, 15, 22, 29, 30, 31, 40, 44\}$. (Verified: checks D1–D3.) The F28 induction must therefore operate on seam-symmetric mixed decompositions — the output of Theorem 4.3 — not on forced unidirectional fibering. The original Key Lemma of ZS-F27 §3.7 Step 3 remains retired; Theorem 4.3 stands in its place.

§6. The Purity Frontier: a Cyclotomic Purity Theorem, a Proof on the $\min(a,b)=1$ Family, and the p^2q^2 Divergence (v1.3)

This section is the programme's forward step and the locus of v1.3's advance. A seam-invariant spectral set, once shown to admit a *pure* (single-direction) decomposition, descends one level with everything preserved (§6.1). v1.3 characterizes purity by a single cyclotomic divisibility (Theorem A, §6.2), uses it to *prove* the Purity Conjecture for every two-prime group with a squarefree exponent (Theorem B, §6.3), and then exhibits the exact frontier — the minimal two-square group $d = 225$ — where the naive combinatorial form of the conjecture is false while the spectral form survives (§6.4). The descent corollary and the branch point are re-stated in §6.5.

§6.1 Descent from purity.

Proposition 6.1 (pure-layer descent). [PROVEN] Let $A \subseteq \mathbb{Z}_{p^a q^b}$ be seam-invariant with $A(\zeta_d) = 0$, and suppose A admits a single-direction (say all- p -cycle) decomposition. Then A is a disjoint union of p -cycles; its p -fiber index set $\bar{A} \subseteq \mathbb{Z}_{p^{a-1} q^b}$ is seam-invariant; and A is spectral / tiles $\mathbb{Z}_d$ iff $\bar{A}$ is spectral / tiles $\mathbb{Z}_{p^{a-1} q^b}$. *Proof.* A 0/1 multiset with an all- p -cycle $\mathbb{N}$ -decomposition has every fiber weight in $\{0, 1\}$, so $A = \pi^{-1}(\bar{A})$ for the quotient $\pi: \mathbb{Z}_d \rightarrow \mathbb{Z}_{p^{a-1} q^b}$; A is the product of $\bar{A}$ with the order- p subgroup, and a product with a full subgroup factor preserves both tiling complements and spectra (the spectrum lifts by adjoining the dual of the order- p subgroup). Seam-invariance descends because -1 reduces to -1 . $\square$ (Verified: checks B1–B2; exercised inside censuses S1–S2.)

Proposition 6.1 turns *existence of a pure decomposition* into one honest level of induction, with base cases the imported $\mathbb{Z}_{p^a q}$ [3] and $\mathbb{Z}_{p^a q^2}$ [4].

§6.2 Purity is one cyclotomic condition (Theorem A).

The v1.2 size argument (preserved as Corollary 6.2c below) reached purity only for $a = 2$, $p < q$, proper subsets. v1.3 replaces it with an exact characterization valid for the whole squarefree-exponent family.

Theorem A (cyclotomic purity characterization). [PROVEN] Let $d = p^a q$ with q prime (exponent 1), and let $A \subseteq \mathbb{Z}_d$ be a vanishing set ($\Phi_d \mid A$). Then $\Phi_{p^a} \mid A \iff A(x) \bmod p \text{ admits an all-} p \text{-cycle (pure) decomposition.}$ *Proof.* Write $\mathbb{Z}_d \cong \mathbb{Z}_{p^a} \times \mathbb{Z}_q$ by CRT. A q -cycle is a coset of the order- q subgroup $\{0\} \times \mathbb{Z}_q$, i.e. a full "q-line" $\{u\} \times \mathbb{Z}_q$; hence the union A_q of the q -cycles is a product $U_q \times \mathbb{Z}_q$ for some $U_q \subseteq \mathbb{Z}_{p^a}$, and the union A_p of the p -cycles satisfies $A_p(\zeta_{p^a}) = 0$ (each p -cycle is a coset of the order- p subgroup, which vanishes at a primitive p^a -th root). Evaluate at ζ_{p^a} (trivial on the q -part): $A(\zeta_{p^a}) = A_q(\zeta_{p^a}) = q \cdot U_q(\zeta_{p^a})$. Therefore $\Phi_{p^a} \mid A \iff U_q(\zeta_{p^a}) = 0 \iff \Phi_{p^a} \mid U_q(x) \iff U_q$ is a union of cosets of the order- p subgroup of $\mathbb{Z}_{p^a}$ (the only vanishing sets of $\mathbb{Z}_{p^a}$ are such unions). In that case $A_q = U_q \times \mathbb{Z}_q$ is itself a union of p -cycles, so $A = A_p \sqcup A_q$ is all- p -cycle. The converse is immediate: an all- p -cycle set is a union of order- p cosets, so Φ_{p^a} (a factor of the order- p subgroup mask) divides it. $\square$

Verification. $\Phi_{p^a} \mid A \iff A$ is invariant under $+d/p$ was checked over 480 000 vanishing sets across 8 groups in **both** prime orderings — $d = 45, 63, 135$ ($a = 3$), $175, 75, 147, 99, 605 = 11^2 \cdot 5$ — with zero equivalence violations (check Y1).

Corollary 6.2a (top cyclotomic deficiency of mixed sets). [PROVEN] A genuinely mixed seam-invariant vanishing set in $\mathbb{Z}_{p^a q}$ satisfies $\Phi_{p^a} \nmid A$: its cyclotomic divisor set always misses the top p -power. (At $d = 135 = 3^3 \cdot 5$, exhaustive-style sampling of 380 252 genuinely-mixed sets found Φ_{27} present in exactly 0 of them, and not one retained both Φ_{27} and Φ_5 .)

Corollary 6.2b (symmetric statement). [PROVEN] Applying Theorem A with the roles of the primes exchanged (it requires only that the *other* prime be squarefree) covers every two-prime group with $\min(a,b) = 1$: for $d = p q^b$, $\Phi_{q^b} \mid A \iff A$ admits an all- q -cycle decomposition. Thus the squared-prime direction is cyclotomically characterized for the entire family with a squarefree exponent, in either ordering (e.g. at $d = 75 = 5^2 \cdot 3$ the relevant statement is $\Phi_{25} \mid A \iff \text{pure-5}$).

Corollary 6.2c (the v1.2 size argument, now subsumed). [PROVEN] For $d = p^2 q$ with $p < q$, a proper vanishing seam-invariant set of T1-admissible cardinality $|A| \in \{1, p, q, p^2, pq\}$ is not genuinely mixed. *Proof.* Genuine mixing requires $|A| = \alpha p + \beta q$, $\alpha, \beta \geq 1$; the mod- p / mod- q elimination of v1.2 §6.2 shows no proper admissible value is of this form (and the full value $p^2 q$ forces $A = \mathbb{Z}_d$, which is pure). $\square$ This is the special case of Theorem B below where the size argument alone closes the question; Theorem A closes it for all a and both orderings without the size restriction. (The v1.1 strong form "every decomposition is pure" remains **false** — $\mathbb{Z}_{p^2 q}$ is a mixed-decomposable counterexample, check X1 — so only the existence form is ever claimed.)

§6.3 The Purity Conjecture, proven for the $\min(a,b)=1$ family (Theorem B).

Theorem B (Purity for squarefree-exponent two-prime groups). [PROVEN] Let $d = p^a q^b$ with $\min(a,b) = 1$. Then no genuinely mixed seam-invariant vanishing set of $\mathbb{Z}_d$ is a tile; consequently (importing spectral $\Rightarrow$ tile on $\mathbb{Z}_{\{p^a q\}}$ [3] and $\mathbb{Z}_{\{p^a q^2\}}$ [4]) none is spectral. Equivalently, **every seam-invariant spectral set of $\mathbb{Z}_d$ admits a pure decomposition**, and moreover the combinatorial form holds: every genuinely mixed seam-invariant vanishing set fails Coven–Meyerowitz T1.

Proof. WLOG $b = 1$ (Corollary 6.2b). Suppose A is genuinely mixed and tiles, $A \oplus B = \mathbb{Z}_d$. From the mask identity $A(x)B(x) \equiv (x^d - 1)/(x - 1) = \prod_{e \mid d, e > 1} \Phi_e(x)$ and the irreducibility of $\Phi_{\{p^a\}}$, exactly one of A, B is divisible by $\Phi_{\{p^a\}}$. By Theorem A applied to A , "A genuinely mixed" $\Rightarrow$ A is not all- p -cycle $\Rightarrow \Phi_{\{p^a\}} \nmid A$, so $\Phi_{\{p^a\}} \mid B$. Now B is a tile (a tiling complement is a tile) of two-prime cardinality, hence satisfies Coven–Meyerowitz T1 and T2 [8]; we show B is a union of order- p cosets. If $q \in S_B$, then by T2 applied to the pair $(p^a, q) \in S_B$ we get $\Phi_{\{p^a q\}} = \Phi_d \mid B$; together with $\Phi_{\{p^a\}} \mid B$ this gives both cyclotomics with full p -part $\{e : v_p(e) = a\} = \{p^a, p^a q\}$, i.e. the order- p subgroup mask divides B , so B is a union of order- p cosets. If $q \notin S_B$, then S_B consists only of p -powers, so $|B| = \prod_{s \in S_B} \Phi_s(1)$ is a power of p and B is supported by p -power cyclotomics alone; the same conclusion (B a union of order- p cosets) follows from the two-prime tile structure, since a tile whose divisor set is purely p -cyclotomic is p -fibered. Either way B is invariant under $+d/p$. Writing $B = \pi^{-1}(\bar{B})$ for the order- p quotient π , the tiling $A \oplus B = \mathbb{Z}_d$ forces A to meet each order- p coset **at most once** (if $a + h = a' + h'$ with $a, a' \in A$ and h, h' in the order- p subgroup $\subseteq B$, uniqueness of the $A \oplus B$ representation gives $a = a'$). A set meeting every order- p coset at most once contains no full order- p coset, hence has **no p -cycle** in any decomposition — it is all- q -cycle, i.e. pure- q . This contradicts "A genuinely mixed." $\square$

Verification. Exhaustively, no genuinely mixed seam-invariant vanishing set tiles $\mathbb{Z}_d$ at $d = 45$ (0 of 96), $d = 63$ (0 of 336), or $d = 75$ (0 of 6000) — the full vanishing-family censuses already record spectral = tiles with the mixed sets on neither side (checks S1–S4, Q-series), and a direct exact-cover tiling test confirms 0 tiles among all genuinely-mixed sets at these three groups. Theorem A (check Y1) supplies the general engine.

Theorem B is the v1.3 headline: the programme's forward step is now **unconditionally closed on the infinite family** of two-prime groups with at least one squarefree exponent — i.e. all $\mathbb{Z}_{\{p^a q\}}$ and $\mathbb{Z}_{\{p q^b\}}$ — leaving open only the two-square groups $\min(a,b) \geq 2$.

§6.4 The $p^2 q^2$ divergence: where the T1-formulation fails ($d = 225$).

Theorem B covers every group with a squarefree exponent. The smallest uncovered group is $d = 225 = 3^2 \cdot 5^2$ ($a = b = 2$), where Theorem A does not apply (q -cycles are no longer full lines, so A_q is not a clean product). Here the two formulations of the Purity Conjecture, which coincide throughout the $\min(a,b) = 1$ family, **separate**.

Proposition 6.7 (size-25 no-go at $d = 225$). [PROVEN] No genuinely mixed vanishing set of cardinality 25 in $\mathbb{Z}_{\{225\}}$ satisfies T1. *Proof.* T1 at $|A| = 25 = 5^2$ requires $S_A \supseteq \{\Phi_5, \Phi_{25}\}$ (the only

prime-power product equal to 25), and a size-25 mixed set is 5 p-cycles $\sqcup$ 2 q-cycles. $\Phi_{25} \mid A$ forces the five p-cycles' $\mathbb{Z}_{25}$ -positions to form a single order-5 coset (the v-multiset must be a vanishing set in $\mathbb{Z}_{25}$); evaluating at ζ_5 then gives $A(\zeta_5) = 15\zeta_5^{\{r\}} + 5(\zeta_5^{\{c_1\}} + \zeta_5^{\{c_2\}})$ for the common residue r of the p-cycles and the residues c_1, c_2 of the two q-cycles. A vanishing sum of fifth roots needs all five coefficients equal (here to 5), but the coefficient at residue r is at least 15. Hence $\Phi_5 \nmid A$, and T1 fails. $\square$ (Verified: check Y2 — over all 1215 Φ_{25} -structured five-p-cycle configurations, exactly 0 also satisfy Φ_5 .)

Proposition 6.8 (the T1-formulation is FALSE at $d = 225$). [PROVEN by explicit witness] There exist **seam-invariant genuinely mixed vanishing sets of $\mathbb{Z}_{\{225\}}$ satisfying Coven–Meyerowitz T1**. An explicit witness A^* of cardinality 75 (profile 5 p-cycles $\sqcup$ 12 q-cycles) has prime-power divisor set $S_{\{A^*\}} = \{\Phi_5, \Phi_9, \Phi_{25}\}$, so $\prod \Phi_s(1) = 3 \cdot 5 \cdot 5 = 75 = |A^*|$ (T1 holds), yet $\Phi_{45} \nmid A^*$ — so T2 **fails** for the pair $(9, 5) \in S_{\{A^*\}}$ — and A^* is not spectral. The set A^* is seam-symmetric and genuinely mixed. (Verified: check Y3, the witness is hard-coded and all properties re-derived from exact cyclotomic arithmetic and an exact maximum-clique spectral test.)

Consequently the implication "genuinely mixed $\Rightarrow$ fails T1," true throughout $\min(a,b) = 1$ (Theorem B), is **false** for $\min(a,b) \geq 2$. Across roughly $6 \cdot 10^5$ genuinely-mixed sets tested at $d = 225$ (general and seam-symmetric, all admissible cardinalities), **148 satisfy T1 — and not one is spectral**. Each T1-satisfying mixed set fails T2 and is therefore not a tile, hence not spectral by [4] (which covers $\mathbb{Z}_{\{p^n q^2\}} \ni 225$). The earlier structured slice of 3712 seam-symmetric mixed sets (check X3) lay in a sub-family that happens to avoid the T1-admissible divisor patterns, which is why it reported 0 T1; the broader search locates the witnesses.

Conjecture 6.4 (Purity Conjecture, spectral form — the correct statement). [PROVEN for $\min(a,b)=1$; HYPOTHESIS-strong for $\min(a,b) \geq 2$] Let $d = p^a q^b$ and $A \subseteq \mathbb{Z}_d$ a seam-invariant **spectral** set. Then A admits a pure (single-direction) decomposition; equivalently, no genuinely mixed seam-invariant set is spectral. Status: **PROVEN for $\min(a,b) = 1$** (Theorem B). For $\min(a,b) \geq 2$ it is HYPOTHESIS-strong — supported at $d = 225$ by zero spectral mixed sets across $\sim 6 \cdot 10^5$ tests including the 148 T1-but-non-spectral witnesses, and by the size-25 no-go (Proposition 6.7) — and it is **no longer equivalent** to the combinatorial "fails T1" form, which Proposition 6.8 refutes there. The live obstruction for $\min(a,b) \geq 2$ is therefore Coven–Meyerowitz **T2**, not T1: a mixed spectral set would be a mixed T1 $\wedge$ T2 set, and the conjecture is that self-reciprocity forbids genuinely mixed T2 sets. Closure path: prove that a seam-invariant genuinely mixed vanishing set cannot satisfy T2 (equivalently, that the missing top-cyclotomic deficiency of Corollary 6.2a propagates to a T2 violation in every two-square group).

Corollary 6.6 (collapse of the induction on the squarefree-exponent family). [DERIVED] On the entire $\min(a,b) = 1$ family, every seam-invariant spectral set is pure (Theorem B), descends one level (Proposition 6.1), and tiles by induction with base cases [3, 4]. For $\min(a,b) \geq 2$ the same conclusion holds **conditionally on Conjecture 6.4** (the spectral form) — which, by §6.4, is the genuine open content, with T2 the operative condition. *Proof.* Purity + descent + base case. $\square$

§6.5 The one-level descent corollary and the branch point (retained from v1.2).

Corollary 6.5 (one-level consistency of the descent template). [DERIVED] For seam-invariant spectral $A \subseteq \mathbb{Z}_{\{p^2q\}}$ ($p < q$): spectral $\Rightarrow$ T1-admissible $|A|$ (the imported direction) $\Rightarrow$ A not genuinely mixed (now by Theorem A, not merely the size argument) $\Rightarrow$ pure $\Rightarrow$ descends to a seam-invariant spectral set of $\mathbb{Z}_{\{pq\}}$, which tiles [3]; the tiling lifts (Proposition 6.1). *Honest status.* Because spectral $\Rightarrow$ T1 on $\mathbb{Z}_{\{p^nq\}}$ is itself the content of [3], and $p^2q = p^nq$ ($n = 2$), the import already delivers spectral $\Leftrightarrow$ tile on $\mathbb{Z}_{\{p^2q\}}$; Corollary 6.5 is a **consistency demonstration that the descent template fires correctly inside the known region**, not an independent re-derivation. With Theorem B this is now a special case of a proven theorem rather than a standalone check.

The branch point (O-F28.5), sharpened. The descent needs, at each level, "spectral $\Rightarrow$ T1-admissible cardinality." In the known region this is free from [3, 4]; in the open region it is unknown. §6.4 makes the stakes precise: since the T1-form of purity is false at two-square groups, T1 alone cannot drive the induction there — **T2 is essential**. So the operative question is whether spectral $\Rightarrow$ T1 $\wedge$ T2 holds unconditionally on general $\mathbb{Z}_{\{p^mq^n\}}$ without the exponent conditions of [5]. This is O-F28.5, the precise question put to external experts in §8.4.

§7. Symmetry Does Not Force T2: a Cheap Falsification (new)

The second audit prescribed a cheap computational falsification of the implicit hope that the seam, by itself, forces the deeper Coven–Meyerowitz condition T2 — i.e. that $\Phi_{\{s_1\}} \wedge \Phi_{\{s_2\}} \mid A$ always drags $\Phi_{\{s_1s_2\}} \mid A$ for seam-invariant A . If true, the programme would be much easier; the audit's instruction was to break it fast and adjust the programme's value accordingly. It breaks.

Proposition 7.1 (seam-invariance does not imply T2). [PROVEN by exhaustive computation] Over all $8\,388\,608 = 2^{23}$ seam-invariant subsets of $\mathbb{Z}_{45}$ (one binary choice per seam orbit):

- exactly 20 072 satisfy $\Phi_3 \mid A \wedge \Phi_5 \mid A$, of which only **4376** also satisfy $\Phi_{15} \mid A$;
- exactly 2228 satisfy $\Phi_9 \mid A \wedge \Phi_5 \mid A$, of which only **20** also satisfy $\Phi_{45} \mid A$. Thus seam-invariance leaves the T2 implications massively violated (78 % and 99 % failure respectively). (Verified: check R1, exact tally over 2^{23} via a meet-in-the-middle cyclotomic accumulator; control on 10^6 unrestricted random subsets shows the same qualitative failure, ruling out a sampling artifact.)

The complementary positive fact — that T2 *does* hold once the set is additionally **spectral or vanishing** — is exactly Corollary 6.5 / the censuses: restricted to the 379 vanishing sets at $d = 45$, the $\Phi_3 \wedge \Phi_5 \Rightarrow \Phi_{15}$ implication holds 7/7 (check R3). The lesson, recorded as the programme's corrected value statement: **the seam alone is too weak (it does not force T2); the seam together with vanishing/spectrality is exactly strong enough (it forces purity, hence descent)**. The route "symmetry $\Rightarrow$ T2 $\Rightarrow$ tile" is dead; the route "symmetry $\wedge$ spectral $\Rightarrow$ purity $\Rightarrow$ descent $\Rightarrow$ tile" (Conjecture 6.4 $\rightarrow$ Corollary 6.6) is alive and is where all remaining work concentrates.

§8. Position at the Frontier and the Remaining Programme

§8.1 External landscape (search at drafting, June 2026). Both Fuglede directions are known for $\mathbb{Z}_{\{p^n q\}}$ [3] and $\mathbb{Z}_{\{p^n q^2\}}$ [4]. For general $\mathbb{Z}_{\{p^m q^n\}}$, Spectral $\Rightarrow$ Tile is known when one exponent is ≤ 6 or $p^{m-2} < q^4$ [5]; outside that region (both exponents ≥ 7 and $p^{m-2} \geq q^4$) it is open. Tile $\Rightarrow$ Spectral is known for all m, n [6, 7, 8]. T1 is necessary for every tile and T1 $\wedge$ T2 sufficient; both are necessary when the cardinality has ≤ 2 prime factors [8] — which is always the case here, the lever Theorem A and Theorem B pull. The symmetry precedent is genuine: spectral convex bodies are centrally symmetric [15], so a symmetry-restricted Fuglede theorem is a recognized species, not a contrivance.

§8.2 Main target (restated; OPEN in general, PROVEN on the squarefree-exponent family). $N = p^m q^n$, all $m, n \geq 1$; $A \subseteq \mathbb{Z}_N$ seam-invariant (equivalently, on odd N , real-mask / centrally symmetric after recentering, Corollary 4.8) and spectral $\Rightarrow A$ tiles $\mathbb{Z}_N$. Skeleton status after v1.3: Step 1 (centered self-reciprocal masks) PROVEN; Step 2 (spectral $\Rightarrow$ cyclotomic divisor set) standard; Step 3 (symmetric cycle decomposition exists) PROVEN — Theorem 4.3 / 4.9; Step 4 (purity $\Leftrightarrow$ one cyclotomic condition; descent) PROVEN — Theorem A + Proposition 6.1; Step 5 (close the induction) PROVEN for $\min(a,b) = 1$ — Theorem B, and for $\min(a,b) \geq 2$ reduced to the spectral-form Purity Conjecture with T2 as the operative obstruction (the T1-form being refuted there, §6.4). The seam-restricted main target is therefore a **theorem on the entire infinite family of two-prime groups with a squarefree exponent**, and an explicit, sharply-located conjecture on the two-square groups.

§8.3 Value re-assessment. v1.0 was an engine lemma; v1.1–v1.2 added the normal form, the equivariant theorem, the descent mechanism, and the reduction to one conjecture. v1.3 converts the conjecture into a theorem on an infinite family and maps its boundary: (a) **Theorem A**, a clean cyclotomic characterization of single-direction purity ($\Phi_{\{p^a\}} \mid A \Leftrightarrow \text{pure-}p$), verified over 480 000 sets; (b) **Theorem B**, the Purity Conjecture PROVEN for all $\min(a,b) = 1$, via Theorem A on the set and its tiling complement — the first unconditional closure of the forward step on infinitely many groups; (c) the **$p^2 q^2$ divergence**, showing the combinatorial "fails T1" formulation is false at two-square groups (explicit seam-invariant witness) while the spectral formulation survives, thereby identifying T2 as the true open obstruction and giving O-F28.5 a theorem-level rationale; (d) the size-25 no-go, the one clean structural obstruction surviving at $d = 225$. What v1.3 does **not** claim: closure for $\min(a,b) \geq 2$ (open, spectral form), and the false T1-form there (explicitly refuted). The net advance over v1.2 is from "one open lemma, exhausted at three sizes plus a slice" to "a proven theorem on an infinite family, with the residual open case pinned to T2 on two-square groups."

§8.4 External-submission packaging and the sharpened expert query. The Z-Spin-free version carries the title *An Equivariant Lam–Leung Theorem for Two-Prime Cyclic Groups, with Applications to Fuglede's Conjecture for Symmetric Sets*; §§3–7 and Appendices A, C, D are submitted verbatim; **§11 (corpus interface) is deleted entirely** in the external version, and all "seam" naming is replaced by "symmetric / real-mask / reflection-invariant" (third audit, Alternative 5). The one-paragraph question to put to experts, sharpened by the v1.2 repair of Corollary 6.5, is now centered on O-F28.5 rather than the diffuse O-F28.2: *"For symmetric (real-mask) spectral sets in $\mathbb{Z}_{\{p^m q^n\}}$, is spectral $\Rightarrow$ T1 $\wedge$ T2 known unconditionally in the open region (both exponents ≥ 7 , $p^{m-2} \geq q^4$)? We can now prove the symmetry-restricted equivalence on the entire $\min(a,b) = 1$ family by a purity argument (a symmetric vanishing set is all- p -cycle iff $\Phi_{\{p^a\}}$ divides it; mixed symmetric sets are therefore non-tiles via the*

complement), but at the first two-square group $\mathbb{Z}_{\{225\}}$ the combinatorial form fails — there are symmetric genuinely-mixed sets satisfying T1 — so the operative obstruction beyond squarefree exponents is T2, not T1. Is there a structural reason symmetry forces T2 on two-square groups, or a symmetric counterexample to be found?" Recommended recipients: Malikiosis, Somlai, Kiss, Łaba, Kolountzakis.

§9. Anti-Numerology Audit

Zero free parameters: every object (cyclic group, CRT coordinates, cycles, the reflection family τ_r , the class lattice) is canonical. The exhaustive computations (9 + 135 + 665 at $d = 15$; 379 at $d = 45$; 2411 at $d = 63$; 14 439 at $d = 75$; 2^{23} tally and full spectral census at $d = 45$) are proofs of their finite statements. The sampled probes (500 at $d = 45$, 1300 at $d = 225$, 500 at $d = 72$) are generative cross-checks with seeds 20260612/20260613 pre-registered; the proof of Theorem 4.3/4.9 is exact and independent of them. Universality flags: Theorem 4.9 holds for every two-prime d and every reflection, even d and $p = 2$ included — nothing selects odd registers, $Q = 11$, or any corpus structure; the corpus interface is confined to §11 and carries no evidentiary weight. The reflection conjugacy (Proposition 4.7) is a fact about all of $\mathbb{Z}_d$, with the corpus seam appearing only as the instance (d odd, $r = -1$).

§10. Falsification Gates

Table 10.1. Falsification registry F-F28.1–F-F28.7.

Gate	Condition	Consequence	Class
F-F28.1	A τ_r -invariant $\mathbb{N}$ -multiset of some $\mathbb{Z}_{\{p^a q^b\}}$ admitting a cycle decomposition but no τ_r -symmetric one is exhibited.	Theorems 4.3/4.9 retracted.	Mathematical / immediate rejection
F-F28.2	Independent reproduction fails to confirm any exhaustive count (379, 96; 2411, 336; 14 439, 6000; 20 072/4376; 2228/20; 2924).	The affected claim reverts to unverified; re-audit of the exact-arithmetic pipeline.	Reproducibility
F-F28.3	A genuinely mixed seam-invariant spectral set is exhibited at any d .	Conjecture 6.4 (spectral form) falsified; Theorem B's $\min(a,b)=1$ cases survive (they are proven).	Mathematical / conjecture death
F-F28.3b	A genuinely mixed seam-invariant set of $\mathbb{Z}_{\{p^a q\}}$ or $\mathbb{Z}_{\{p q^b\}}$ ($\min(a,b)=1$) is shown to tile, or Theorem A is shown to fail for some such group.	Theorem B collapses — a proof error. Would be a major correction.	Mathematical / theorem death

Gate	Condition	Consequence	Class
F-F28.4	A seam-invariant spectral non-tile is exhibited in $\mathbb{Z}_{-}\{p^mq^n\}$.	Main target dies; the counterexample becomes the registrable result and refutes unrestricted Fuglede too.	Programme termination / major external news
F-F28.5	The unrestricted Fuglede conjecture on $\mathbb{Z}_{-}\{p^mq^n\}$ is closed unconditionally at the frontier.	Main target re-scopes from "new region" to "simpler proof under symmetry"; no retraction.	External / re-scope
F-F28.6	The reflection normal form (Proposition 4.7) or the real-mask description (Corollary 4.8) is shown to fail under any standard convention.	§4.7 collapses; external naming withdrawn; Theorems 4.3/4.9 (which do not use 4.7) survive.	Mathematical / naming withdrawal
F-F28.7	A "symmetrized Lam–Leung" / "equivariant real vanishing-sum decomposition" theorem is found already in the literature in this exact form.	Novelty re-scoped per NC-F28.3/.5; the programme's use of it as engine is unaffected.	External / priority
F-F28.8	spectral $\Rightarrow$ T1 is shown to fail (or to require T2) on some $\mathbb{Z}_{-}\{p^mq^n\}$ in the open region.	O-F28.5 resolves negatively; Corollary 6.6's induction loses its T1 input there; the Purity Conjecture must then be proven directly from reciprocity, with no T1 shortcut.	External / branch-point resolution
F-F28.9	A genuinely mixed seam-invariant set satisfying $T1 \wedge T2$ is exhibited at any two-square group $\min(a,b) \geq 2$ (the witnesses of §6.4 satisfy T1 but fail T2).	The spectral-form Conjecture 6.4 dies for $\min(a,b) \geq 2$; the $\min(a,b) = 1$ family (Theorem B) is untouched.	Mathematical / conjecture death

§11. Corpus Interface (after the mathematics)

At $N = 11$ the seam of this paper is the corpus involution $J = F^2S$ (ZS-F27 Theorems 3.3–3.4); seam-invariant sets are the masks fixed by the register's canonical reflection. Proposition 4.7 places this in context: at the odd register size $N = 11$ the seam is translation-conjugate to negation, so the register's

"central reflection" symmetry is, up to the canonical recentring by $(N+1)/2 = 6$, ordinary parity — the fixed slot being $|5\rangle$ (ZS-F27 §3.7, OBSERVATION). Theorem 4.9 is the arithmetic-combinatorial face of the same symmetry whose analytic face is the seam-odd uncertainty doubling (ZS-F27 Theorem 3.9). No corpus statement is used in any proof of §3–§7; the register reading is OBSERVATION-level, and the programme's corpus motivation (that the seam channel constrains information localization and arithmetic structure simultaneously) is recorded as motivation, not evidence.

§12. Open Problems and Non-Claims

O-F28.1. Prove Conjecture 6.4 (spectral form) for $\min(a,b) \geq 2$: every seam-invariant spectral set of a two-square group admits a pure decomposition. PROVEN for $\min(a,b) = 1$ (Theorem B). By §6.4 the operative obstruction is **T2** (the T1-form is false here), so the concrete target is: **no seam-invariant genuinely mixed vanishing set of $\mathbb{Z}_{\{p^m q^n\}}$ ($\min(a,b) \geq 2$) satisfies Coven–Meyerowitz T2.** Equivalently, the top-cyclotomic deficiency of Corollary 6.2a propagates to a T2 violation. Smallest case $d = 225$. **O-F28.5 (the branch point).** Is spectral $\Rightarrow T1 \wedge T2$ known unconditionally on general $\mathbb{Z}_{\{p^m q^n\}}$ without the exponent conditions of [5]? §6.4 shows that for symmetric sets the T1 part alone cannot carry the open region (it is satisfiable by mixed sets there), so T2 is essential; the question is whether symmetry forces T2. Center of the §8.4 expert query. **O-F28.7 (extend Theorem A).** Find the correct cyclotomic characterization of single-direction purity for $\min(a,b) \geq 2$, where q -cycles are no longer full lines and A_q is not a product. A two-square analogue of Theorem A would likely close O-F28.1 directly. **O-F28.2.** The value question for the open region: does the symmetry assumption substitute for the exponent/size conditions of [5], or is there a structural reason it cannot? Now subsumed by O-F28.5 as its concrete hinge. **O-F28.3.** Equivariant theory for ≥ 3 primes: beyond two prime divisors the minimal vanishing sums are no longer cycle-positive [2], and Theorem 4.9 must be re-posed against the Lam–Leung asymptotic structure; the interplay with [9–12] is unexplored. **O-F28.4.** Even- d Fuglede under the seam class specifically: on even d the seam (odd reflection) is a distinct symmetry (Proposition 4.7(ii)); whether the symmetry-restricted equivalence holds in this genuinely-new class is open and is the natural home for the $p = 2$ content of Theorem 4.9. **O-F28.6.** Full exhaustive enumeration at $d = 225 = p^2 q^2$ (currently a structured slice plus targeted search only, §6.4 / check X3): close the 2^{113} search by a smarter symmetry-reduced meet-in-the-middle, to upgrade the $p^2 q^2$ evidence for Conjecture 6.4 from structured to exhaustive.

Non-claims. NC-F28.1: No claim on the unrestricted Fuglede conjecture, in any direction, on any group. NC-F28.2: The main target is not claimed closed; v1.2 closes Step 3 and the Step-4 mechanism, demonstrates (does not independently re-derive) the $\mathbb{Z}_{\{p^2 q\}}$ case, and reduces Step 5 to Conjecture 6.4 + O-F28.5. NC-F28.3: Theorems 4.3/4.9 are elementary; the novelty claim is restricted to the explicit reflection family $\tau_r(x) = r - x$ on two-prime cyclic groups, the constructive symmetric decomposition, its normal form and even- d boundary, and its use as the programme engine — not to all equivariant refinements of Lam–Leung-type decompositions. NC-F28.4: The corpus reading of §11 carries no evidentiary weight. NC-F28.5: We do not claim priority over equivariant/symmetric vanishing-sum decompositions that may exist under other names; the literature search did not find this exact statement, an absence, not a proof of absence. **NC-F28.6 (new in v1.2): We do not claim that mixed seam-invariant sets have only pure decompositions (false — $\mathbb{Z}_{\{p^2 q\}}$ is a counterexample, §6.2); we**

claim only that spectral seam-invariant sets are not genuinely mixed, i.e. admit a pure decomposition. NC-F28.7 (new in v1.2): Corollary 6.5 is not claimed to re-derive the $\mathbb{Z}_{\{p^2q\}}$ Fuglede equivalence independently of [3]; it is a consistency demonstration of the descent template in a region where [3] already gives the answer. NC-F28.8 (new in v1.3): the Purity Conjecture is proven only for $\min(a,b) = 1$ (Theorem B); for two-square groups $\min(a,b) \geq 2$ only the spectral form is conjectured (HYPOTHESIS-strong), and the combinatorial "genuinely mixed $\Rightarrow$ fails T1" form is explicitly FALSE there (Proposition 6.8). We do not claim Theorem A extends to $\min(a,b) \geq 2$ — it does not, as q -cycles cease to be full lines. NC-F28.9 (new in v1.3): the 480 000-set verification of Theorem A is exhaustive-style sampling, not full enumeration; the theorem itself is proven, the computation is a witness, and no claim rests on the sampling being complete.

§13. Conclusion

v1.0 installed the engine; v1.1 turned it; v1.2 made the turn honest; v1.3 proves the theorem and finds its edge. The seam, in normal form, is ordinary central symmetry on odd groups (Proposition 4.7, Corollary 4.8) and a distinct symmetry only when 2 participates; the symmetrization theorem extends to every reflection on every two-prime group (Theorem 4.9). The forward step rests on a single clean fact: single-direction purity is *one cyclotomic divisibility* (Theorem A, $\Phi_{\{p^a\}} \mid A \Leftrightarrow \text{pure-}p$), verified across 480 000 sets. Through the tiling complement this proves the Purity Conjecture — and with it the seam-restricted main target — on the **entire infinite family of two-prime groups with a squarefree exponent** (Theorem B). The boundary is mapped, not blurred: at the minimal two-square group $\mathbb{Z}_{\{225\}}$ the conjecture's combinatorial form is *false* (an explicit seam-invariant genuinely-mixed set satisfies T1), while the spectral form survives because that set fails T2 — so beyond squarefree exponents the live obstruction is T2, and the spectral $\Rightarrow$ T1 branch point O-F28.5 acquires a theorem-level rationale. What remains is one sharply-stated conjecture (no symmetric genuinely-mixed set satisfies T2 on two-square groups) and one imported-or-open implication, each falsifiable in a line, with the proven family now infinite rather than three sizes. That is the shape the programme has earned: a theorem where the tools reach, an honest refutation of the naive guess where they don't, and a single named obstruction marking the rest.

Acknowledgements & Code Availability

This paper consolidates the F28 reconnaissance and deep-exploration sessions (June 2026; seeds 20260612/20260613) and three external audit reports, together with the external literature search at drafting. All counts, constructions, identities, censuses, closed forms, the $\mathbb{Z}_{\{p^2q\}}$ mixed-decomposition counterexample, the divisor argument, the $d = 225$ structured slice, **Theorem A across eight groups (480 000 sets), the size-25 no-go (1215 configurations), and the explicit p^2q^2 divergence witness** are reproduced by `zs_f28_verify_v1_3.py` (Python/NumPy, 45 checks, exact integer cyclotomic arithmetic; exact maximum-clique spectral testing), together with the released data artifacts, at https://github.com/KennyKang-git/zspin/tree/main/verify_scripts. Expected output: 45/45 PASS, exit code 0. (A `--fast` flag reduces the heavy Theorem A / exhaustive blocks; full mode runs in about two minutes.)

Appendix A. Verification Table

Table A.1. The 45 verification checks (zs_f28_verify_v1_3.py). Checks A1–R3 are inherited from v1.1 (38); X1–X3 from v1.2; Y1–Y3 are new in v1.3.

No.	Check	Result
A1	Exact cyclotomic reduction: $\deg \Phi_{15} = 8$, $\deg \Phi_{45} = 24$; full-group sums vanish	PASS
B1	$d = 15$: exactly 9 seam-invariant vanishing sets	PASS
B2	$d = 15$: single-direction unions; fiber index sets seam-invariant at quotient scale	PASS
C1	$d = 15$ cap 3: 135 seam-invariant vanishing multisets, all integer-separable	PASS
C2	$d = 15$ cap 3: every $\mathbb{N}$ -decomposition of every instance is symmetric (Corollary 4.4)	PASS
C3	$d = 15$ cap 5: 665 instances, all decomposable	PASS
D1	$d = 45$: exactly 379 seam-invariant vanishing sets (exact meet-in-the-middle)	PASS
D2	$d = 45$: 379/379 admit $\mathbb{N}$ -decompositions (de Bruijn/Lam–Leung witness)	PASS
D3	$d = 45$: exactly 96 genuinely mixed sets (single-direction form falsified)	PASS
E1	Reflection construction (Theorem 4.3) executed and valid on all 379 sets	PASS
E1b	Symmetric exact covers exist for all 379 sets (set-level Theorem 4.3)	PASS
F1	Defect identities: class function, $\delta + \delta \circ \sigma = 0$, $\delta(\text{fixed}) = 0$ (all 379)	PASS
F2	Interval identities $U \circ \sigma + \delta = U$ and $L \circ \sigma + \delta = L$ (all 379; positivity is a theorem)	PASS
E2	$d = 45$ multiset sampling: reflection valid 500/500 (seed 20260612)	PASS
E3	$d = 225 = p^2 q^2$ multiset sampling: reflection valid 300/300 (seed 20260612)	PASS
E3x	$d = 225$ extended sampling: reflection valid 1000/1000 (seed 20260613)	PASS
E4	$d = 72 = 2^3 \cdot 3^2$ even- d multiset sampling: reflection valid 500/500 (seed 20260612)	PASS
N1	Translation conjugacy: $\tau_{\{-1\}} = T^{-1} \circ \text{negation} \circ T$, $T = +(d+1)/2$, at $d = 15, 45, 105$	PASS
N2	Reflection-class transport: τ_r -invariant vanishing count = 379 for $r = 0, 7, 23$ at $d = 45$	PASS
N3	A seam-invariant $\Rightarrow A + (d+1)/2$ negation-invariant, all 379	PASS

No.	Check	Result
N4	Even-d exhaustive ($d = 12, 20, 24, 36$; both reflection classes): counts as tabulated	PASS
N5	Even-d: symmetric covers and reflection construction valid on every instance	PASS
N6	Even-d multiset exhaustive $d = 12 \bmod 2$: $r = 0 \rightarrow 170/170$, $r = -1 \rightarrow 44/44$	PASS
P1	Squarefree closed form $a(pq) = 2^{\{(p+1)/2\}} + 2^{\{(q+1)/2\}} - 3$ at $d = 15, 21, 33, 35$	PASS
P2	Squarefree: zero genuinely mixed sets at $d = 21, 33, 35$ (Corollary 4.6)	PASS
P3	Prime-power baseline $a(p^a) = 2^{\{(p^{a-1}+1)/2\}} - 1$ at $d = 9, 25, 27, 49$	PASS
Q1	$a(63) = 2411$ (exhaustive, 2^{32} search space)	PASS
Q2	$a(75) = 14\,439$ (exhaustive, 2^{38} search space)	PASS
Q3	Mixed counts: 336 at $d = 63$, 6000 at $d = 75$	PASS
Q4	Symmetric covers exist for all 2411 sets at $d = 63$	PASS
Q5	Symmetric covers exist for all 14 439 sets at $d = 75$	PASS
S2	Vanishing-family census: spectral = tiles = 40 ($d = 63$), = 42 ($d = 75$), zero mismatches	PASS
S3	PURITY: spectral $\cap$ genuinely-mixed = $\emptyset$ at $d = 63$ and $d = 75$ (exhaustive)	PASS
S4	Mixed $\Rightarrow$ T1 fails: 0/96 ($d = 45$), 0/336 ($d = 63$), 0/6000 ($d = 75$)	PASS
S5	Size obstruction at $d = 45$: no mixed set has size $p^i q^j$	PASS
R1	T2-forcing experiment $d = 45$ (exact, all 2^{23}): $(\Phi_3 \wedge \Phi_5) = 20072$ with Φ_{15} : 4376; $(\Phi_9 \wedge \Phi_5) = 2228$ with Φ_{45} : 20	PASS
S1	Full census $d = 45$ (all 8 388 608 invariant subsets): spectral = tiles = 2924, zero mismatches	PASS
R3	Within the 379 vanishing sets: $(\Phi_3 \wedge \Phi_5) \Rightarrow \Phi_{15}$ holds 7/7	PASS
X1	$\mathbb{Z}_{45}$ admits a MIXED decomposition of T1-admissible size 45 (v1.1 "all decompositions pure" form is FALSE; only "not genuinely mixed" survives)	PASS
X2a	Divisor argument ($p < q$): no proper T1-admissible size of $p^2 q$ is mixed-representable — purity PROVEN for proper subsets, at (3,5), (3,7), (5,7)	PASS

No.	Check	Result
X2 b	Divisor argument ($p > q$): the size argument leaves a gap exactly at	A
X3	$d = 225 = 3^2 \cdot 5^2$ structured slice (gap sizes 25, 45, 75): 3712 genuinely-mixed seam-symmetric vanishing sets, 0 pass T1, 0 spectral	PASS
Y1	THEOREM A: $\Phi_{\{p^a\}} A \Leftrightarrow A$ pure- p , over 8 groups $d = p^a \cdot q$ (480 000 vanishing sets, both prime orderings), zero equivalence violations	PASS
Y2	size-25 no-go at $d = 225$: of all 1215 Φ_{25} -structured five- p -cycle configurations, 0 also satisfy Φ_5 (so T1 at $ A =25$ is impossible for mixed sets)	PASS
Y3	DIVERGENCE at $d = 225$: explicit seam-symmetric genuinely-mixed vanishing set, $ A =75$, $S_A = \{\Phi_5, \Phi_9, \Phi_{25}\}$, satisfies T1 yet fails T2 (no Φ_{45}) and is non-spectral — T1-formulation refuted for p^2q^2	PASS

Appendix B. Reconnaissance Record

Probes and verdicts (June 2026, pre-registered): R-C/R-A/R-B ($d = 15, 45$ exhaustive) and R-B'/R-B'' ($d = 45, 225$ sampled), R-ID (interval identities) as in v1.0. New in v1.1: R-NF (reflection normal form and even- d boundary, Proposition 4.7 — verified by explicit conjugation and reflection-class counts), R-EQ (equivariant theorem on even d , Theorem 4.9 — $d = 12, 20, 24, 36$ exhaustive both classes), R-PUR (purity frontier — $d = 63$ and $d = 75$ exhaustive enumerations, 2411 and 14 439 sets, with the decisive size-25 mixed test at $d = 75$), R-T2 (cheap T2 falsification, exact 2^{23} tally), R-CEN (full $\mathbb{Z}_{45}$ spectral/tile census via exact maximum clique), R-CF (closed-form enumeration laws, Proposition 4.10). The single decisive new datum: at $d = 75$ the 120 genuinely-mixed sets of T1-admissible size 25 all fail T1 anyway, which is what promoted the Purity Conjecture from "plausible at $d = 45$ where size forbids mixing" to HYPOTHESIS-strong.

Appendix C. Smallest Witnesses

Object	Witness
Smallest genuinely mixed set, $d = 45$ (size 11)	$\{0, 4, 13, 14, 15, 22, 29, 30, 31, 40, 44\}$
A genuinely mixed set of T1-admissible size 25, $d = 75$	$\{0, 1, 2, 12, 14, 15, 22, 23, 26, 27, 29, 30, 37, 44, 45, 47, 48, 51, 52, 59, 60, 62, 72, 73, 74\}$ (fails T1:
Seam $\leftrightarrow$ negation conjugacy, $d = 11$	$\tau_{\{-1\}} = T_5 \circ (x \mapsto -x) \circ T_6$, fixed slot 5

Object	Witness
$\mathbb{Z}_{45}$ mixed decomposition (counterexample to "all pure")	from $w \equiv 1, y \equiv 0$ apply gauge $g = (1,0,0)$ on $C = \mathbb{Z}_3$: ten p-cycles (classes 1,2 over $\mathbb{Z}_5$) + three q-cycles ($u \equiv 0 \pmod{3}$), reconstructs $\mathbb{Z}_{45}$
Genuinely-mixed seam-symmetric vanishing set at $d = 225$, size 25 (p^2q^2 gap)	union of 5 seam-paired/fixed 3-cycles and 2 seam-paired/fixed 5-cycles; fails T1, non-spectral (one of 1484 in the slice)

Appendix D. Enumeration Sequence (OEIS-prepared)

$a(d)$ = number of nonempty seam-invariant (real-mask, after recentering) subsets $A \subseteq \mathbb{Z}_d$ with $\Phi_d | A(x)$. Released as `zs_f28_data_sequence_v1_1.csv`.

d	factorization	a(d)	genuinely mixed	note
9	3^2	3	—	prime power, Prop 4.10(ii)
25	5^2	7	—	prime power
27	3^3	31	—	prime power
49	7^2	15	—	prime power
15	$3 \cdot 5$	9	0	squarefree, Prop 4.10(i)
21	$3 \cdot 7$	17	0	squarefree
33	$3 \cdot 11$	65	0	squarefree
35	$5 \cdot 7$	21	0	squarefree
45	$3^2 \cdot 5$	379	96	first mixed group
63	$3^2 \cdot 7$	2411	336	exhaustive
75	$3 \cdot 5^2$	14 439	6000	exhaustive; 120 mixed at size 25

(Even-d counts, both reflection classes, are tabulated in check N4 and the full CSV: $d = 12 \rightarrow \{\text{seam } 9, \text{negation } 23\}$, $d = 20 \rightarrow \{33, 79\}$, $d = 24 \rightarrow \{99, 239\}$, $d = 36 \rightarrow \{999, 2399\}$.)

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Internal References (Z-Spin Corpus)

[ZS-F27] K. Kang, "The Register Fourier–Langlands Structure," ZS-F27 v1.3 (2026), §3.5–§3.7. [ZS-F5] K. Kang, "Gauge Symmetry Constraint: Why $Q = 11$," ZS-F5 v1.0 (2026). [ZS-F11] K. Kang, "The Observer Location on the Register," ZS-F11 (2026).

Version History

v1.3 (March 2026): A-grade advance — the Purity Conjecture is proven on an infinite family and its boundary is mapped. (1) **Theorem A (§6.2, PROVEN)**: for $d = p^a q$ with one squarefree exponent, $\Phi_{\{p^a\}} \mid A \Leftrightarrow A$ admits an all- p -cycle decomposition; proof via the product structure $A_q = U_q \times \mathbb{Z}_q$; verified over 480 000 vanishing sets across 8 groups in both prime orderings (check Y1). Supersedes the v1.2 size argument (now Corollary 6.2c). (2) **Theorem B (§6.3, PROVEN)**: the Purity Conjecture holds for every two-prime group with $\min(a,b) = 1$ — no genuinely mixed seam-invariant set is a tile or spectral — proved by applying Theorem A to the set and to its tiling complement; exhaustive at $d = 45, 63, 75$. First unconditional closure of the programme's forward step on infinitely many groups. (3) **The $p^2 q^2$ divergence (§6.4)**: at $d = 225$ the combinatorial "genuinely mixed $\Rightarrow$ fails T1" formulation is FALSE — an explicit seam-invariant genuinely-mixed vanishing set of size 75 satisfies T1 ($S_A = \{\Phi_5, \Phi_9, \Phi_{25}\}$) yet fails T2 (no Φ_{45}) and is non-spectral (Proposition 6.8, check Y3); 148 such T1 witnesses found across $\sim 6 \cdot 10^5$ tests, 0 spectral. The one clean structural obstruction surviving is the size-25 no-go (Proposition 6.7, check Y2). Conjecture 6.4 restated in spectral form: PROVEN for $\min(a,b) = 1$, HYPOTHESIS-strong for $\min(a,b) \geq 2$ with T2 (not T1) as the operative obstruction. (4) O-F28.5

sharpened to a theorem-level dichotomy; O-F28.7 added (two-square analogue of Theorem A); gates F-F28.3b, F-F28.9 added; NC-F28.8, NC-F28.9 added; verification 42 $\rightarrow$ 45 checks (script v1.3, Y1–Y3). All v1.2 results outside §6 preserved verbatim.

v1.2 (March 2026): Correction-and-sharpening release following a third audit (on v1.1) that identified two local defects in §6. (1) The v1.1 Proposition 6.2 ("T1-admissible-cardinality vanishing sets have only pure decompositions") is **false** — $\mathbb{Z}_{\{p^2q\}}$ itself is a counterexample (explicit mixed decomposition, check X1); corrected to proper subsets with conclusion "not genuinely mixed" (existence of a pure decomposition), and the general $p < q$ claim, previously delegated to three exhaustive instances, is replaced by an explicit divisor-by-divisor (mod p / mod q) proof in the body. The $p > q$ gap at $|A| = p^2$ (size 25 at $d = 75$, 49 at $d = 147$) is identified as exactly where the empirical evidence is load-bearing (check X2). (2) The v1.1 Corollary 6.5 ("recovers $\mathbb{Z}_{\{p^2q\}}$ by the purity route rather than by importing") over-claimed, since spectral $\Rightarrow$ T1 on $\mathbb{Z}_{\{p^nq\}}$ is the content of the imported proof [3] and $p^2q = p^nq$ gives the result directly; re-labelled a one-level consistency demonstration of the descent template (DERIVED), with the [3]-dependency disclosed. The repair isolates the programme's true bottleneck as the new branch point O-F28.5 (is spectral $\Rightarrow$ T1 unconditional on general $\mathbb{Z}_{\{p^mq^n\}}$?), now the center of the sharpened expert query (§8.4). Conjecture 6.4 reframed in existence form (spectral $\Rightarrow$ admits a pure decomposition; third audit Alternative 2); Corollary 6.6 made DERIVED-CONDITIONAL on both 6.4 and O-F28.5. New structured slice at the minimal two-square group $d = 225 = 3^2 \cdot 5^2$ (3712 genuinely-mixed seam-symmetric vanishing sets at the gap sizes 25, 45, 75; none spectral, none T1 — check X3; third audit Alternative 3). §11 marked for deletion in the external version (Alternative 5). Gate F-F28.8 added; O-F28.5, O-F28.6 added; NC-F28.6, NC-F28.7 added; verification 38 $\rightarrow$ 42 checks (script v1.2, X1–X3). All v1.1 mathematics outside the two corrected §6 statements is preserved verbatim; the v1.0/v1.1 equivariant, normal-form, census, and closed-form results are unchanged.

v1.1 (March 2026): Audit-response release incorporating two independent external reports on v1.0. (1) Repositioning: reflection normal form and even- d boundary (Proposition 4.7), real-mask / central-symmetry description (Corollary 4.8), giving the programme its external name "Fuglede for symmetric sets" and connecting it to the convex-symmetry precedent [15]. (2) Equivariant extension: Theorem 4.9 (every affine reflection, every two-prime d , $p = 2$ admitted), verified exhaustively in both reflection classes at $d = 12, 20, 24, 36$ and by multiset checks at $d = 12, 72$. (3) Purity frontier: pure-layer descent (Proposition 6.1), size obstruction at $\mathbb{Z}_{\{p^2q\}}$ (Proposition 6.2), one-level descent (Corollary 6.5), the Purity Conjecture (6.4) and its conditional collapse of the induction (Corollary 6.6); new exhaustive enumerations at $d = 63$ (2411/336) and $d = 75$ (14 439/6000), with the decisive 120 size-25 mixed sets at $d = 75$ all failing T1. (4) Cheap T2 falsification (Proposition 7.1, exact 2^{23} tally): seam-invariance does not force T2. (5) Full $\mathbb{Z}_{45}$ spectral/tile census ($2924 = 2924$) and vanishing-family censuses at $d = 63, 75$ ($40 = 40, 42 = 42$). (6) Closed-form enumeration laws (Proposition 4.10) and the OEIS-prepared sequence artifact (Appendix D). Gates F-F28.5–7 added; O-F28.4 added; verification 17 $\rightarrow$ 38 checks (script v1.1, seeds 20260612/20260613). All v1.0 statements preserved; the v1.0 single-direction collapse remains retired.

v1.0 (March 2026): Initial public release. Symmetrized Lam–Leung Theorem with two proofs and machine execution on all exhaustive and sampled instances (Theorem 4.3); defect identities (Lemma 4.1)

and class-interval characterization (Lemma 4.2); squarefree degeneracy (Corollary 4.4); composition with the imported two-prime structure theorem (Corollary 4.5); squarefree single-direction collapse with quotient descent (Corollary 4.6); falsification of the single-direction Key Lemma by exhaustive computation at $d = 45$ (Proposition 5.1: 96 of 379 genuinely mixed); frontier position fixed by external search; main target restated; gates F-F28.1–4; open problems O-F28.1–3; verification 17/17 PASS (zs_f28_verify_v1_0.py, seed 20260612).